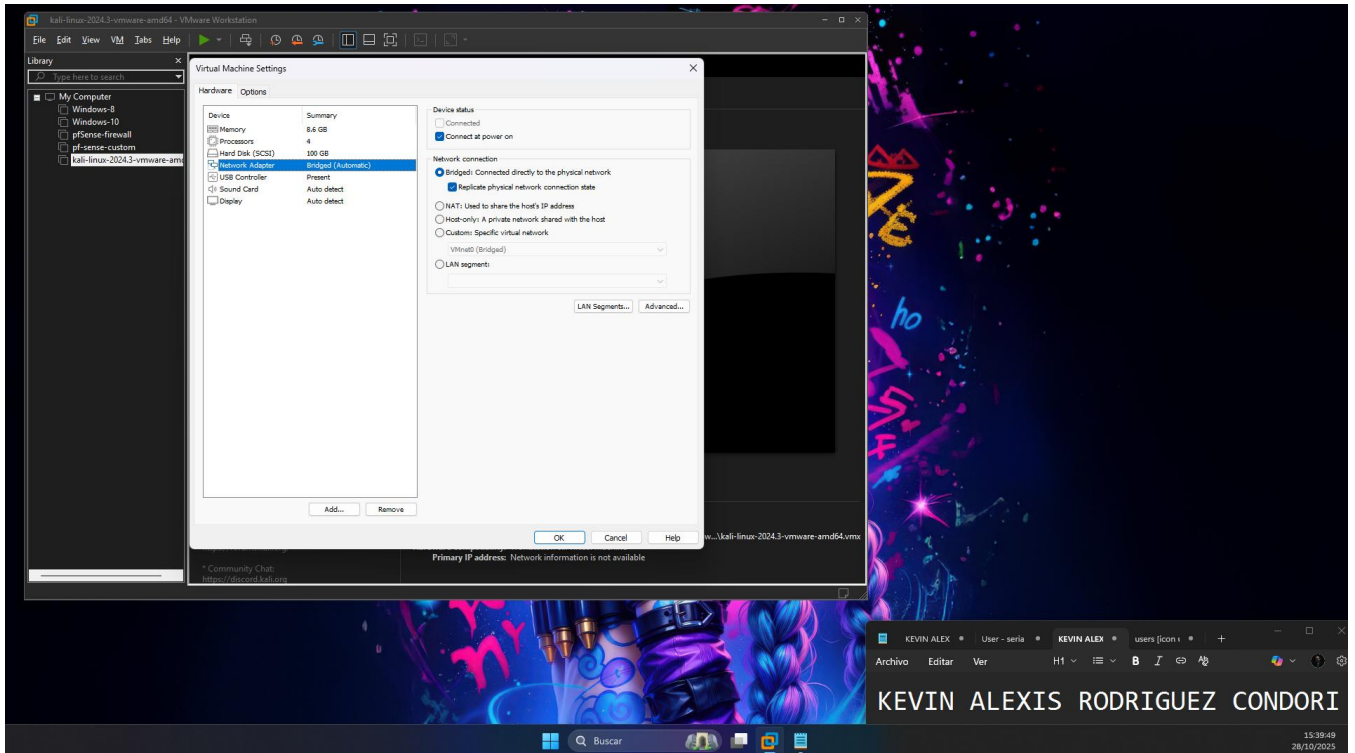


LABORATORIO # 8 – SEGURIDAD DE SISTEMAS

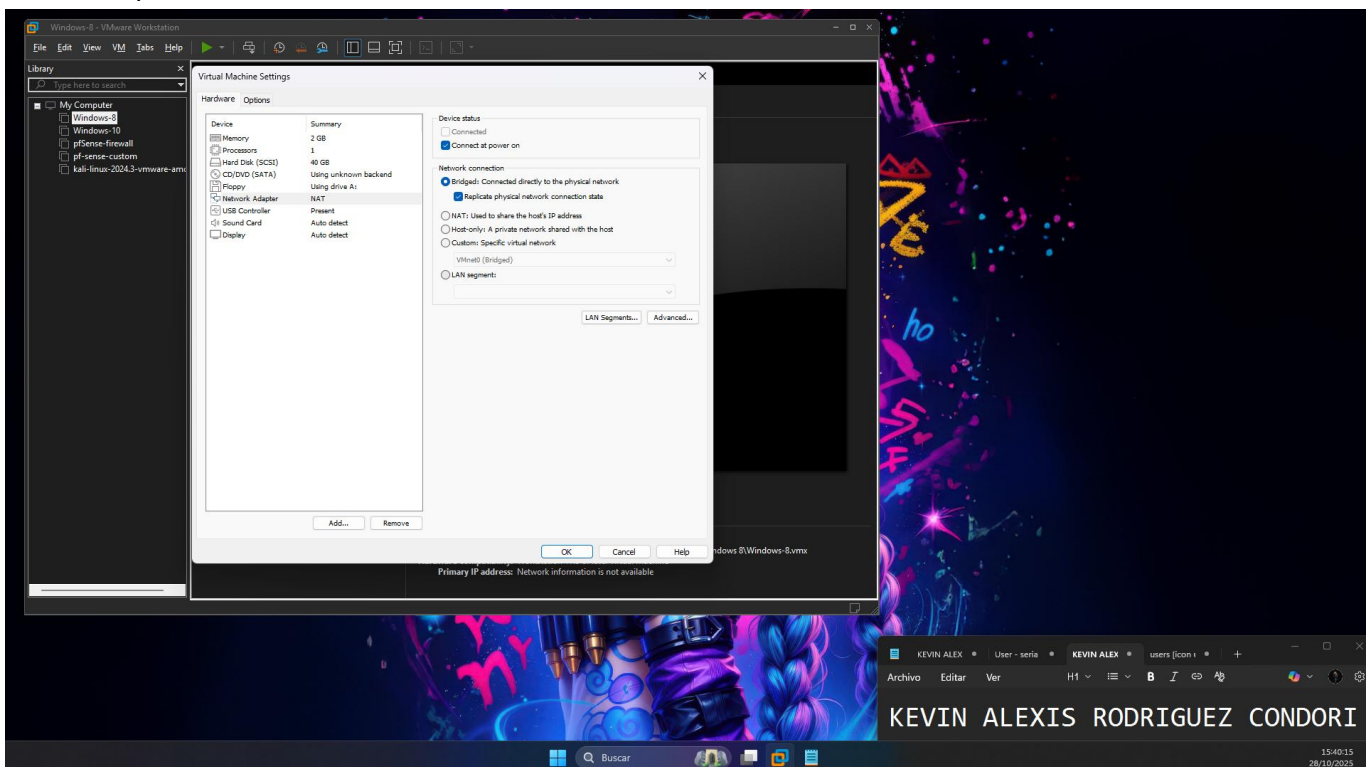
Nombre: Univ. Kevin Alexis Rodríguez Condori

Verificaciones

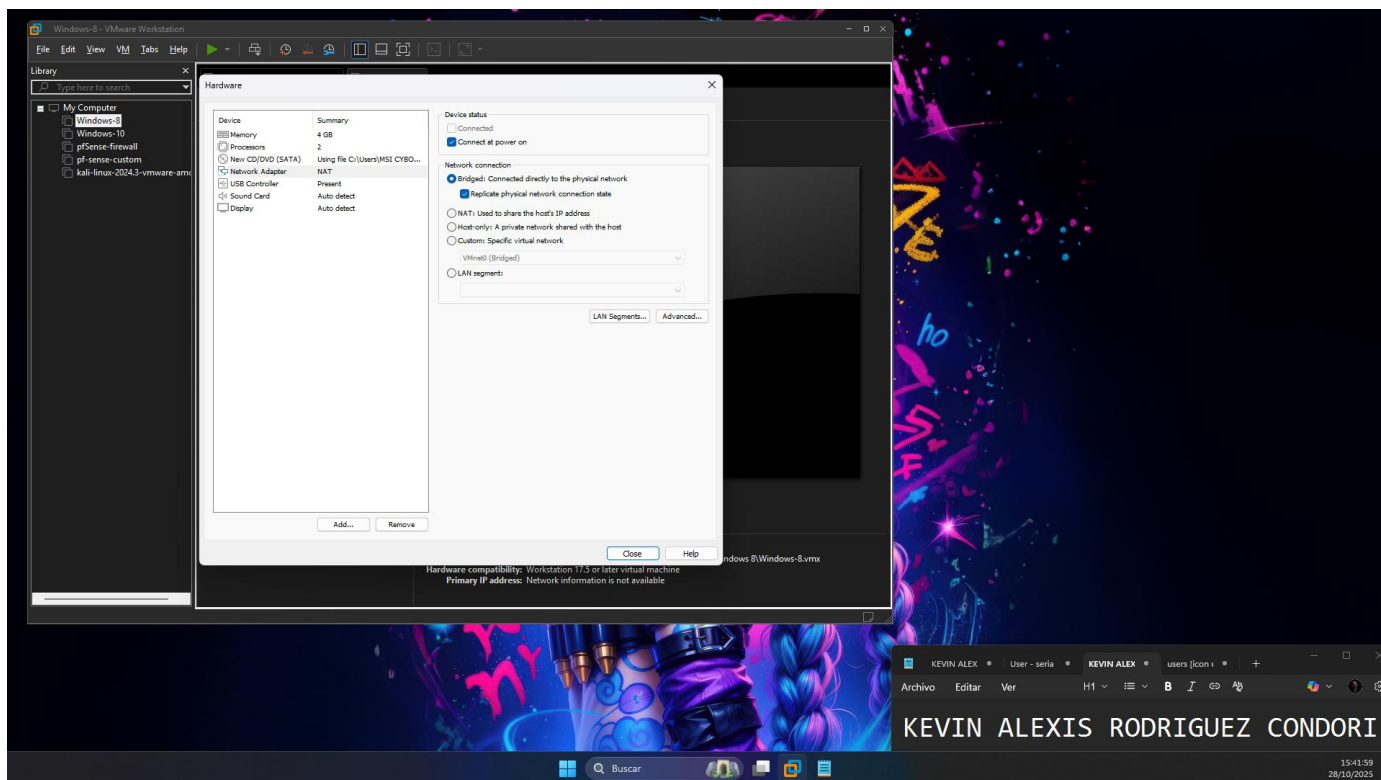
❖ Sistema Operativo Kali Linux



❖ Sistema Operativo Ubuntu 18



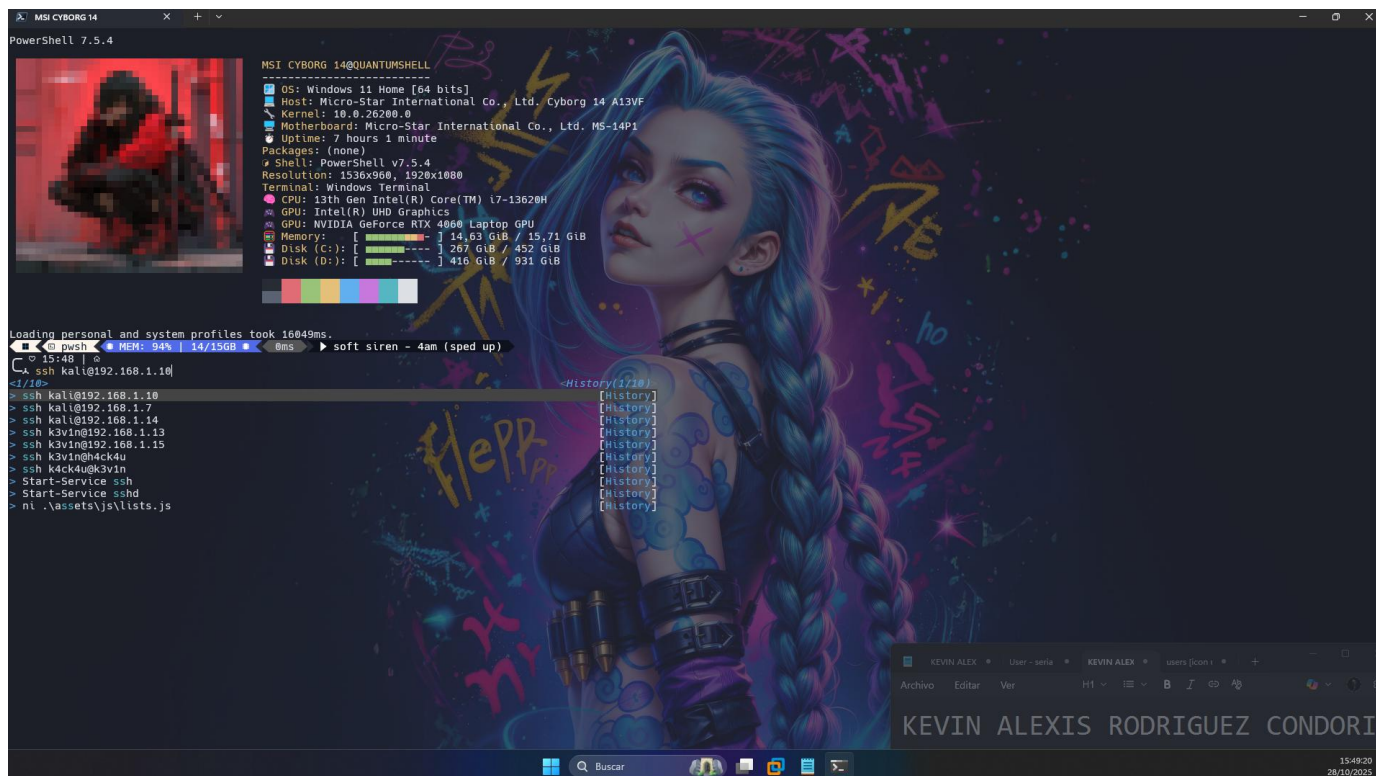
❖ Sistema operativo Windows 8

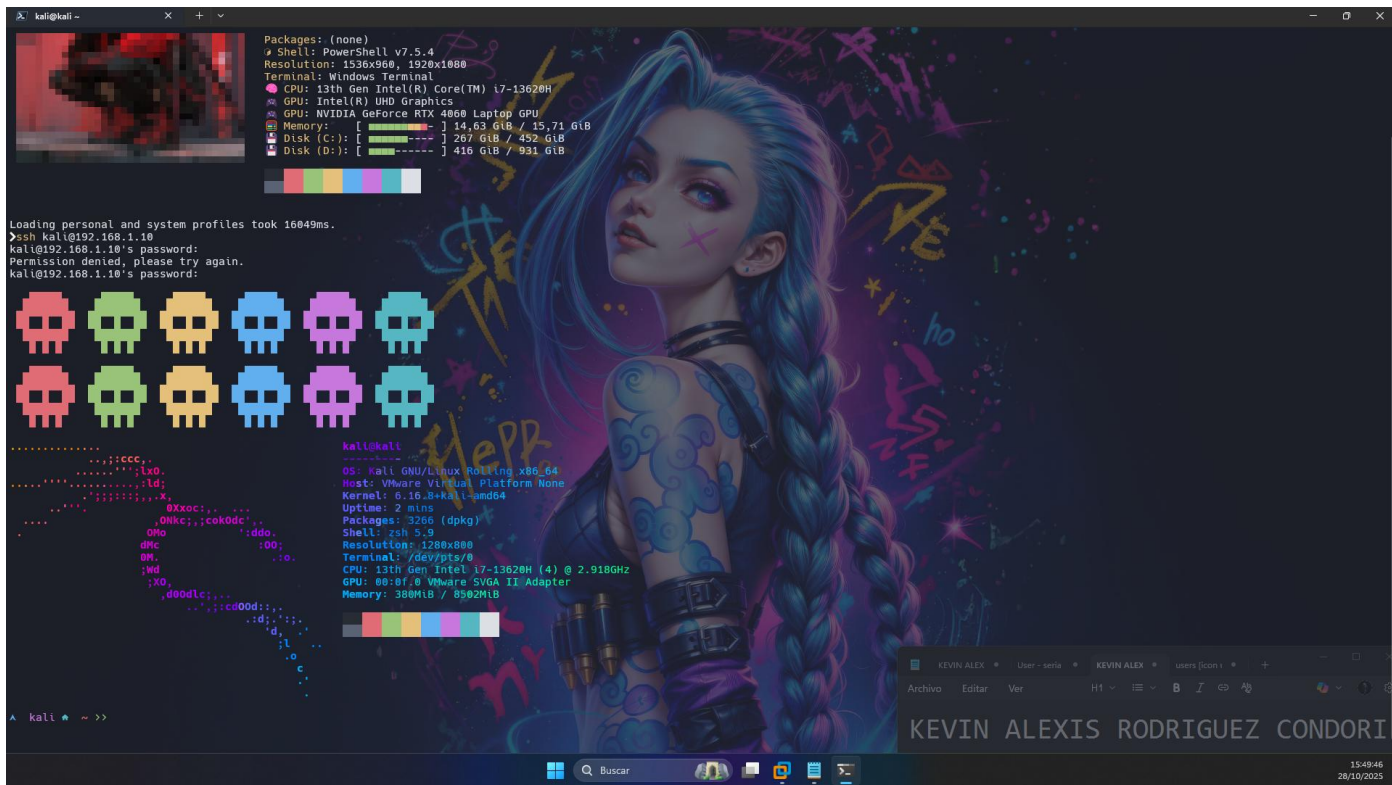


2. Configuración e instalación

Para toda la configuración se usa Kali Linux, entonces iniciamos la VM.

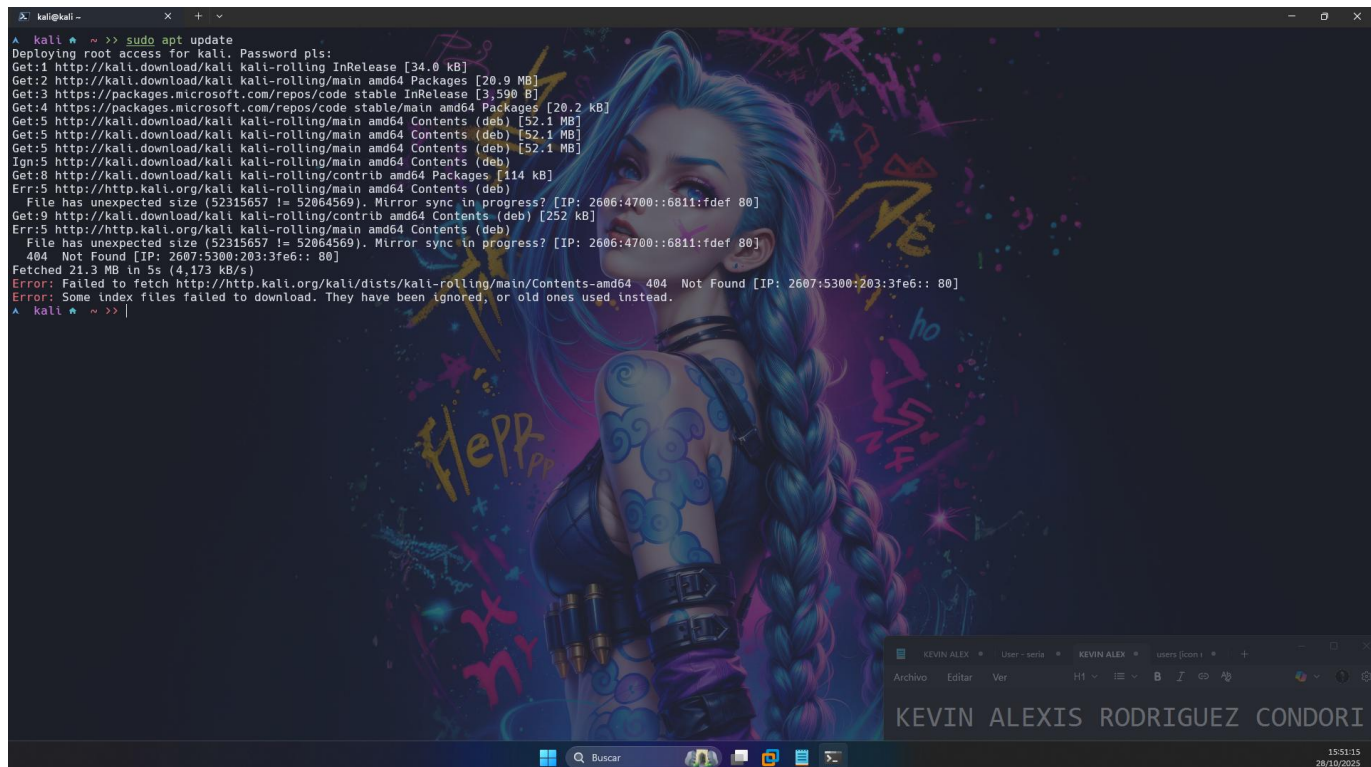
Me conectare por ssh desde la maquina física.





❖ Instalación

Primero antes que nada siempre se recomienda ejecutar los siguientes comandos: `sudo apt update`



Instalamos freeradius


```
kali@kali ~ >> sudo apt update
Deploying root access for kali. Password pls:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Get:4 https://packages.microsoft.com/repos/code stable/main amd64 Packages [20.2 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.1 MB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.1 MB]
Get:5 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.1 MB]
Ign:5 http://kali.download/kali kali-rolling/main amd64 Contents (deb)
Get:8 http://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]
Err:5 http://http.kali.org/kali kali-rolling/main amd64 Contents (deb)
File has unexpected size (52315657 != 52064569). Mirror sync in progress? [IP: 2606:4700::6811:fdef 80]
Get:9 http://http.kali.org/kali kali-rolling/contrib amd64 Contents (deb) [252 kB]
Err:5 http://http.kali.org/kali kali-rolling/main amd64 Contents (deb)
File has unexpected size (52315657 != 52064569). Mirror sync in progress? [IP: 2606:4700::6811:fdef 80]
404 Not Found [IP: 2607:5300:203:3fe6:: 80]
Fetched 21.3 MB in 5s (4,173 kB/s)
Error: Failed to fetch http://http.kali.org/kali/dists/kali-rolling/main/Contents-amd64 404 Not Found [IP: 2607:5300:203:3fe6:: 80]
Error: Some index files failed to download. They have been ignored, or old ones used instead.
kali@kali ~ >> sudo apt install freeradius
```

```
kali@kali ~ >> sudo apt install freeradius
Preparing to unpack .../1-freeradius-config_3.2.8+dfsg-1_amd64.deb ...
Unpacking freeradius-config (3.2.8+dfsg-1) ...
Selecting previously unselected package libfreeradius3.
Preparing to unpack .../2-libfreeradius3_3.2.8+dfsg-1_amd64.deb ...
Unpacking libfreeradius3 (3.2.8+dfsg-1) ...
Selecting previously unselected package freeradius.
Preparing to unpack .../3-freeradius_3.2.8+dfsg-1_amd64.deb ...
Unpacking freeradius (3.2.8+dfsg-1) ...
Selecting previously unselected package freeradius-utils.
Preparing to unpack .../4-freeradius-utils_3.2.8+dfsg-1_amd64.deb ...
Unpacking freeradius-utils (3.2.8+dfsg-1) ...
Selecting previously unselected package libdbi-perl:amd64.
Preparing to unpack .../5-libdbi-perl_1.647-1_amd64.deb ...
Unpacking libdbi-perl:amd64 (1.647-1) ...
Setting up libfreeradius3 (3.2.8+dfsg-1) ...
Setting up freeradius-common (3.2.8+dfsg-1) ...
Setting up libdbi-perl:amd64 (1.647-1) ...
Setting up freeradius-config (3.2.8+dfsg-1) ...
Generating DH parameters, 1024 bit long safe prime
.....
*****
Setting up freeradius (3.2.8+dfsg-1) ...
update-rc.d: We have no instructions for the freeradius init script.
update-rc.d: It looks like a network service, we disable it.
update-rc.d: warning: start and stop actions are no longer supported; falling back to defaults
freeradius.service is a disabled or a static unit, not starting it.
Setting up freeradius-utils (3.2.8+dfsg-1) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

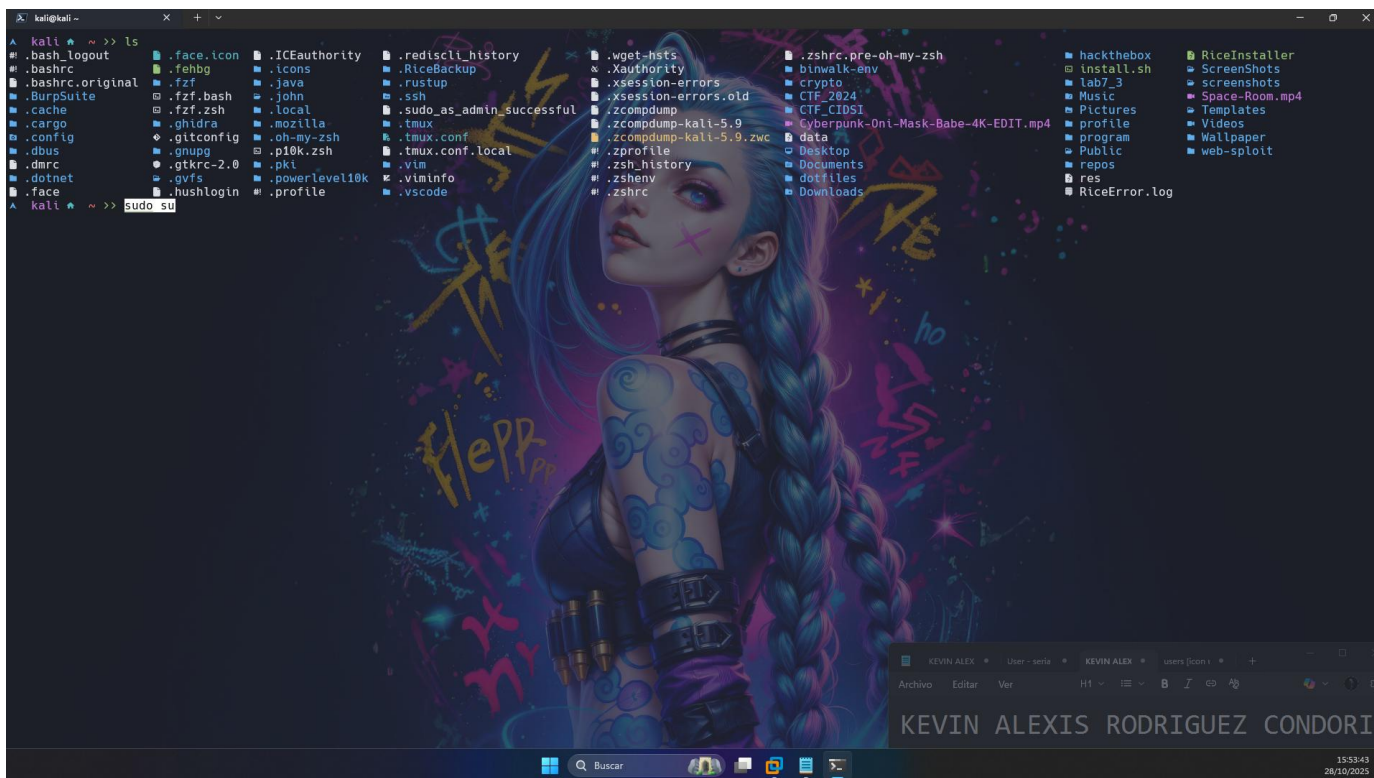
No services need to be restarted.

No containers need to be restarted.

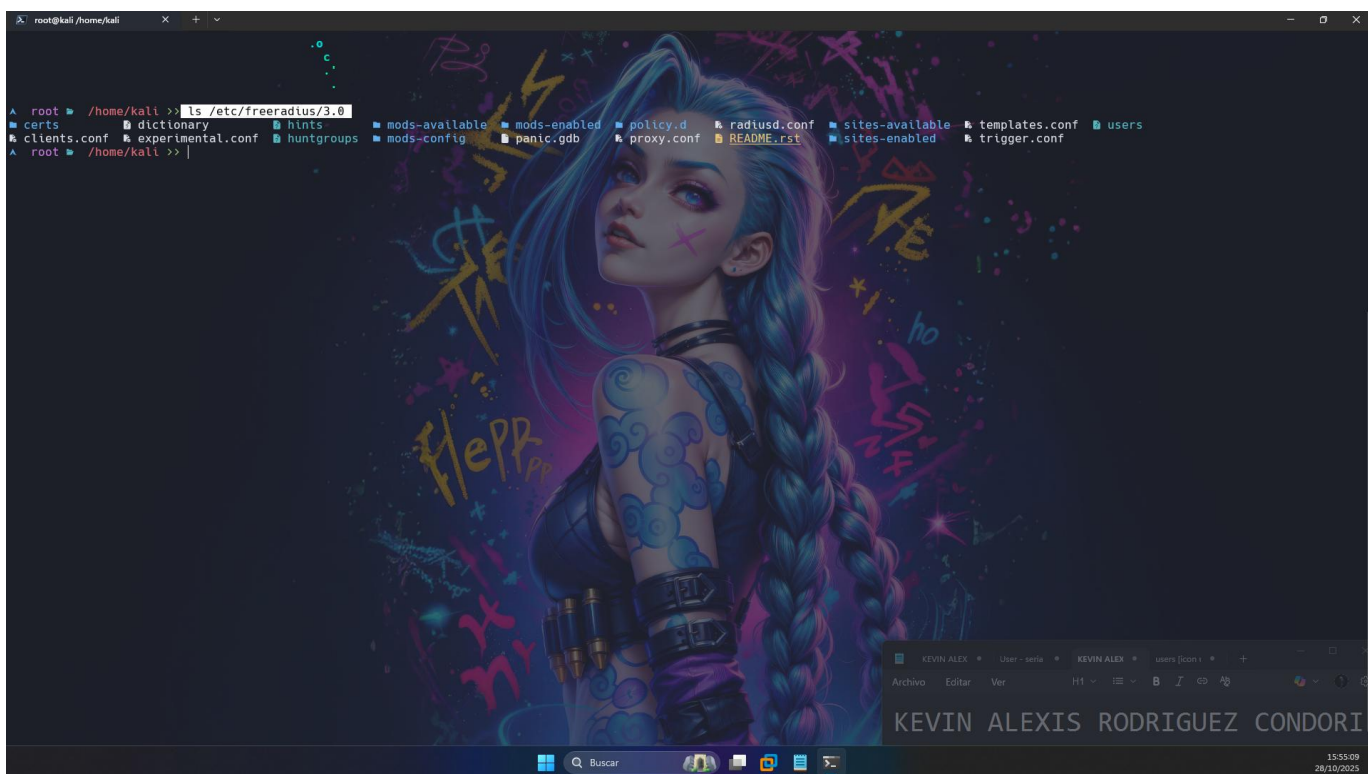
No user sessions are running outdated binaries.

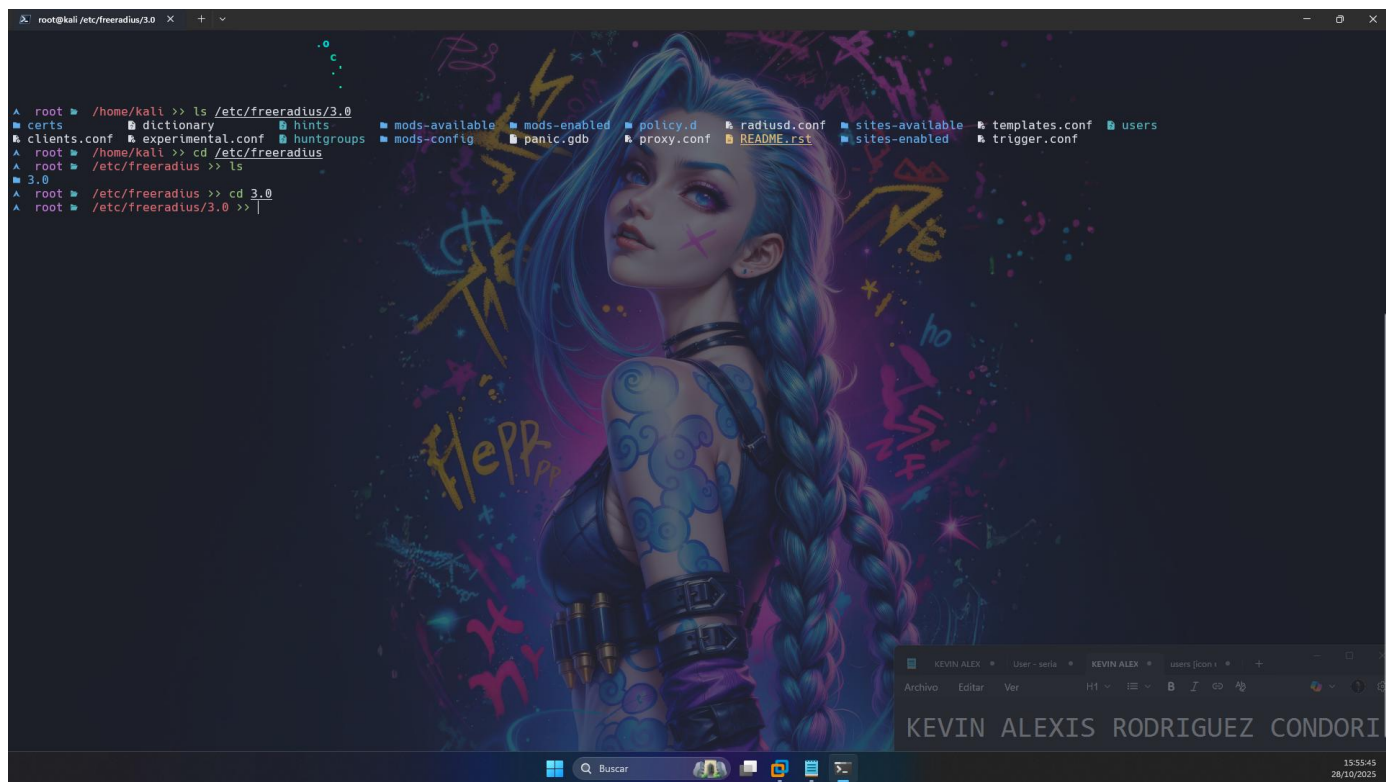
No VM guests are running outdated hypervisor (qemu) binaries on this host.
kali@kali ~ >> |
```

Cambiamos a super usuario

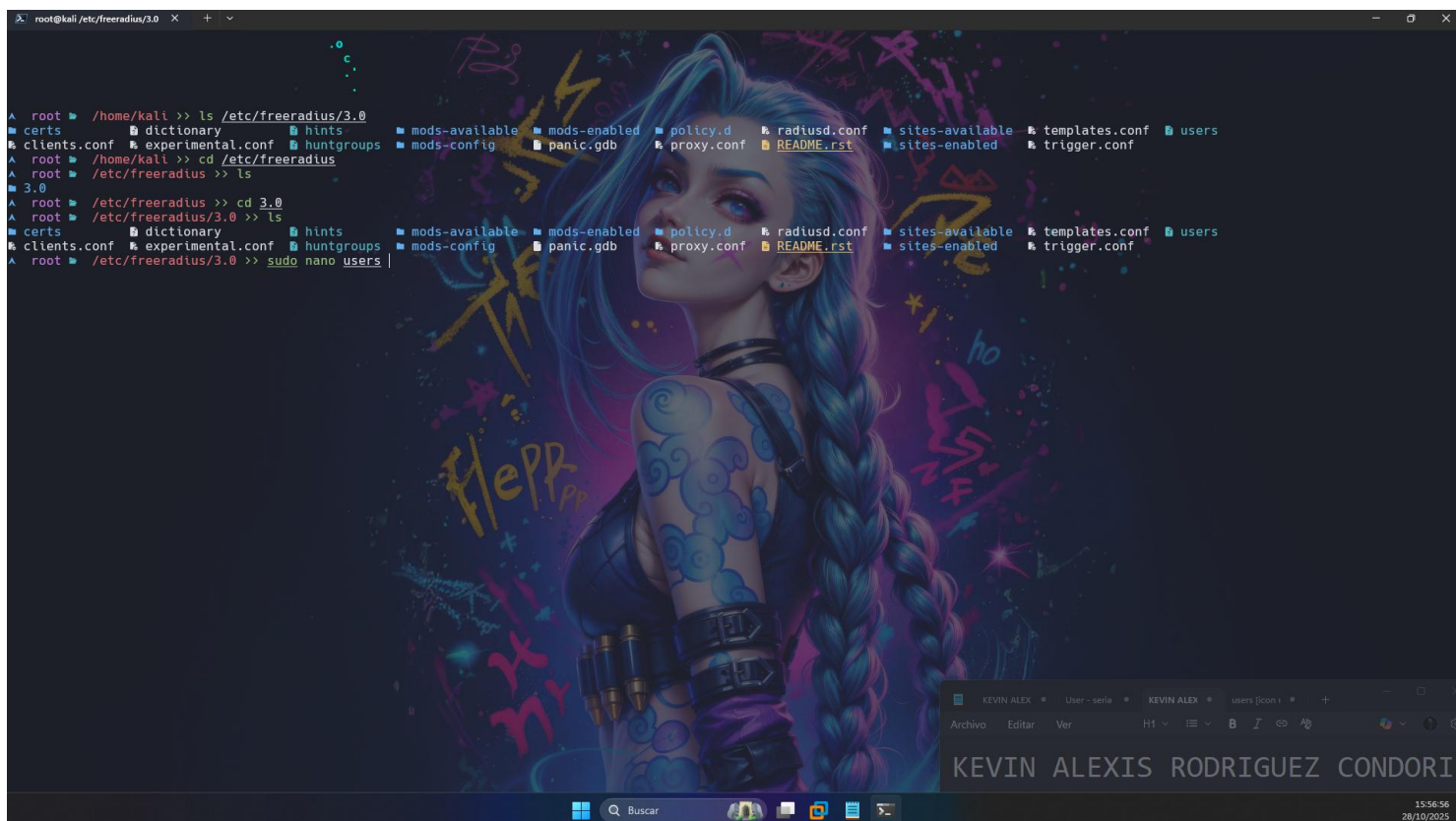


Verificamos los archivos de freeradius





Configuramos usuarios



Configuramos un usuario llamado “k3v1n h4ck” con una contraseña 12345

```
GNU nano 8.6 users *
# NOTE: we do not use Hint = "PPP", since PPP might also be auto-detected
# by the terminal server in which case there may not be a "P" suffix.
# The terminal server sends "Framed-Protocol = PPP" for auto PPP.
#
DEFAULT Framed-Protocol == PPP
Framed-Protocol = PPP,
Framed-Compression = Van-Jacobson-TCP-IP
#
# Default for CSLIP: dynamic IP address, SLIP mode, VJ-compression.
#
DEFAULT Hint == "CSLIP"
Framed-Protocol = SLIP,
Framed-Compression = Van-Jacobson-TCP-IP
#
# Default for SLIP: dynamic IP address, SLIP mode.
#
DEFAULT Hint == "SLIP"
Framed-Protocol = SLIP
#
# Last default: rlogin to our main server.
#
#DEFAULT
# Service-Type = Login-User,
# Login-Service = Rlogin,
# Login-IP-Host = shellbox.ispdomain.com
#
# #
# # Last default: shell on the local terminal server.
# #
# DEFAULT
# Service-Type = Administrative-User
#
# On no match, the user is denied access.
#
#####
# You should add test accounts to the TOP of this file! #
# See the example user "bob" above.
#####
# "John Doe" Cleartext-Password := "hello"
# Reply-Message = "Hello, %{User-Name}"
# "k3v1n h4ck" Cleartext-Password := "12345"

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward
```

Creamos un usuario llamado admin, con una contraseña y un mensaje de bienvenida

```
GNU nano 8.6 users *
# NOTE: we do not use Hint = "PPP", since PPP might also be auto-detected
# by the terminal server in which case there may not be a "P" suffix.
# The terminal server sends "Framed-Protocol = PPP" for auto PPP.
#
DEFAULT Framed-Protocol == PPP
Framed-Protocol = PPP,
Framed-Compression = Van-Jacobson-TCP-IP
#
# Default for CSLIP: dynamic IP address, SLIP mode, VJ-compression.
#
DEFAULT Hint == "CSLIP"
Framed-Protocol = SLIP,
Framed-Compression = Van-Jacobson-TCP-IP
#
# Default for SLIP: dynamic IP address, SLIP mode.
#
DEFAULT Hint == "SLIP"
Framed-Protocol = SLIP
#
# Last default: rlogin to our main server.
#
#DEFAULT
# Service-Type = Login-User,
# Login-Service = Rlogin,
# Login-IP-Host = shellbox.ispdomain.com
#
# #
# # Last default: shell on the local terminal server.
# #
# DEFAULT
# Service-Type = Administrative-User
#
# On no match, the user is denied access.
#
#####
# You should add test accounts to the TOP of this file! #
# See the example user "bob" above.
#####
# "John Doe" Cleartext-Password := "hello"
# Reply-Message = "Hello, %{User-Name}"
# "k3v1n h4ck" Cleartext-Password := "12345"
admin Cleartext-Password := "admin"
Reply-Message = "Hello, %{User-Name}"

Soft wrapping of overlong lines enabled
Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward
```

Configuración de clientes: nano clients.conf, 192.168.1.10


```
GNU nano 8.6 clients.conf *
# the same as above, but they are nested inside of a section.
# You can have as many per-socket client lists as you have "listen"
# sections, or you can re-use a list among multiple "listen" sections.
# Un-comment this section, and edit a "listen" section to add:
# "clients = per_socket_clients". That IP address/port combination
# will then accept ONLY the clients listed in this section.
#
# There are additional considerations when using clients from SQL.
#
# A client can be link to a virtual server via modules such as SQL.
# This link is done via the following process:
#
# If there is no listener in a virtual server, SQL clients are added
# to the global list for that virtual server.
#
# If there is a listener, and the first listener does not have a
# "clients=..." configuration item, SQL clients are added to the
# global list.
#
# If there is a listener, and the first one does have a "clients=..."
# configuration item, SQL clients are added to that list. The client
# { ... } configured in that list are also added for that listener.
#
# The only issue is if you have multiple listeners in a virtual
# server, each with a different client list, then the SQL clients are
# added only to the first listener.
#
#clients per_socket_clients {
#  client socket_client {
#    ipaddr = 192.0.2.4
#    secret = testing123
#  }
#}

client 192.168.1.10 {
  secret = sistemas
  shortname = router1
  nastype = cisco
}

client 0.0.0.0/0 {
  ipaddr = 0.0.0.0/0
  secret = sistemas
  shortname = windows-client
}
```

Habilitamos los logs:

En radiusd.conf habilitamos las siguientes opciones:

auth = yes

auth_badpass = yes

auth_goodpass = yes

```
GNU nano 8.6 radiusd.conf *
# don't want to change this.
#
syslog_facility = daemon

# Log the full User-Name attribute, as it was found in the request.
#
# allowed values: {no, yes}
#
stripped_names = no

# Log all (accept and reject) authentication results to the log file.
#
# This is the same as setting "auth_accept = yes" and
# "auth_reject = yes"
#
# allowed values: {no, yes}
#
auth = yes

# Log Access-Accept results to the log file.
#
# This is only used if "auth = no"
#
# allowed values: {no, yes}
#
auth_accept = no

# Log Access-Reject results to the log file.
#
# This is only used if "auth = no"
#
# allowed values: {no, yes}
#
auth_reject = no

# Log passwords with the authentication requests.
#
# auth_badpass - logs password if it's rejected
# auth_goodpass - logs password if it's correct
#
# allowed values: {no, yes}
#
auth_badpass = yes
auth_goodpass = yes

# Log additional text at the end of the "Login OK" messages.
#
# for those to work, the "auth" and "auth_goodpass" or "auth_badpass"
# configurations above have to be set to "yes".
```


Reiniciamos el servicio de freeradius

```
root@kali:/etc/freeradius/3.0# cat radiusd.conf
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:7d:ba:46 brd ff:ff:ff:ff:ff:ff
inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
    valid_lft 84314sec preferred_lft 84314sec
inet6 2800::cd0:bb3a:9200:6959:15c6:d77f:2e2a/64 scope global dynamic noprefixroute
    valid_lft 2098sec preferred_lft 2092sec
inet6 fe80::58aa:6337:927f:8087/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: br-4074da9bcc16: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:09:6a:db:20 brd ff:ff:ff:ff:ff:ff
inet 172.18.0.1/16 brd 172.18.255.255 scope global br-4074da9bcc16
    valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:f3:0d:c1:41 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
root@kali:/etc/freeradius/3.0# ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:7d:ba:46 brd ff:ff:ff:ff:ff:ff
inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
    valid_lft 84294sec preferred_lft 84294sec
inet6 2800::cd0:bb3a:9200:6959:15c6:d77f:2e2a/64 scope global dynamic noprefixroute
    valid_lft 2079sec preferred_lft 2073sec
inet6 fe80::58aa:6337:927f:8087/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: br-4074da9bcc16: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:09:6a:db:20 brd ff:ff:ff:ff:ff:ff
inet 172.18.0.1/16 brd 172.18.255.255 scope global br-4074da9bcc16
    valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:f3:0d:c1:41 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
root@kali:/etc/freeradius/3.0# nano radiusd.conf
root@kali:/etc/freeradius/3.0# systemctl status freeradius
o freeradius.service - FreeRADIUS multi-protocol policy server
Loaded: loaded (/usr/lib/systemd/system/freeradius.service; disabled; preset: disabled)
Active: inactive (dead)
Docs: man:radiusd(8)
      man:radiusd.conf(5)
      http://wiki.freeradius.org/
      http://networkradius.com/doc/
root@kali:/etc/freeradius/3.0#
```

```
root@kali:/etc/freeradius/3.0# cat radiusd.conf
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:7d:ba:46 brd ff:ff:ff:ff:ff:ff
inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
    valid_lft 84314sec preferred_lft 84314sec
inet6 2800::cd0:bb3a:9200:6959:15c6:d77f:2e2a/64 scope global dynamic noprefixroute
    valid_lft 2098sec preferred_lft 2092sec
inet6 fe80::58aa:6337:927f:8087/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: br-4074da9bcc16: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:09:6a:db:20 brd ff:ff:ff:ff:ff:ff
inet 172.18.0.1/16 brd 172.18.255.255 scope global br-4074da9bcc16
    valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:f3:0d:c1:41 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
root@kali:/etc/freeradius/3.0# ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:7d:ba:46 brd ff:ff:ff:ff:ff:ff
inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
    valid_lft 84294sec preferred_lft 84294sec
inet6 2800::cd0:bb3a:9200:6959:15c6:d77f:2e2a/64 scope global dynamic noprefixroute
    valid_lft 2079sec preferred_lft 2073sec
inet6 fe80::58aa:6337:927f:8087/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: br-4074da9bcc16: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:09:6a:db:20 brd ff:ff:ff:ff:ff:ff
inet 172.18.0.1/16 brd 172.18.255.255 scope global br-4074da9bcc16
    valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:f3:0d:c1:41 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
root@kali:/etc/freeradius/3.0# nano radiusd.conf
root@kali:/etc/freeradius/3.0# systemctl status freeradius
o freeradius.service - FreeRADIUS multi-protocol policy server
Loaded: loaded (/usr/lib/systemd/system/freeradius.service; disabled; preset: disabled)
Active: inactive (dead)
Docs: man:radiusd(8)
      man:radiusd.conf(5)
      http://wiki.freeradius.org/
      http://networkradius.com/doc/
root@kali:/etc/freeradius/3.0# systemctl restart freeradius
root@kali:/etc/freeradius/3.0#
```



```
valid_lft 2079sec preferred_lft 2073sec
inet6 fe80::5baa:6337:927f:8087/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: br-4074da9bcc16: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:09:6a:db:20 brd ff:ff:ff:ff:ff:ff
inet 172.18.0.1/16 brd 172.18.255.255 scope global br-4074da9bcc16
valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
link/ether 02:42:f3:0d:c1:11 brd ff:ff:ff:ff:ff:ff
inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0

root@kali:/etc/freeradius/3.0# nano radiusd.conf
root@kali:/etc/freeradius/3.0# systemctl status freeradius
○ freeradius.service - FreeRADIUS multi-protocol policy server
Loaded: loaded (/usr/lib/systemd/system/freeradius.service; disabled; preset: disabled)
Active: inactive (dead)
Docs: man:radiusd(8)
      man:radiusd.conf(5)
      http://wiki.freeradius.org/
      http://networkradius.com/doc/
root@kali:/etc/freeradius/3.0# systemctl restart freeradius
root@kali:/etc/freeradius/3.0# systemctl status freeradius
● freeradius.service - FreeRADIUS multi-protocol policy server
Loaded: loaded (/usr/lib/systemd/system/freeradius.service; disabled; preset: disabled)
Active: active (running) since Tue 2025-10-28 16:36:20 -04; 20s ago
Invocation: c39c4b59665948678ef20055a10ec7
Docs: man:radiusd(8)
      man:radiusd.conf(5)
      http://wiki.freeradius.org/
      http://networkradius.com/doc/
Process: 5125 ExecStartPre=/usr/sbin/freeradius $FREERADIUS_OPTIONS -Cx -lstdout (code=exited, status=0/SUCCESS)
Main PID: 5130 (freeradius)
Status: "Processing requests"
Tasks: 6 (limit: 9956)
Memory: 43.2M (max: 2G, available: 1.9G, peak: 45.3M)
CPU: 30ms
CGroup: /system.slice/freeradius.service
└─5130 /usr/sbin/freeradius -f

Oct 28 16:36:20 kali freeradius[5125]: Compiling Auth-Type PAP for attr Auth-Type
Oct 28 16:36:20 kali freeradius[5125]: Compiling Auth-Type CHAP for attr Auth-Type
Oct 28 16:36:20 kali freeradius[5125]: Compiling Auth-Type MS-CHAP for attr Auth-Type
Oct 28 16:36:20 kali freeradius[5125]: Compiling Autz-Type New-TLS-Connection for attr Autz-Type
Oct 28 16:36:20 kali freeradius[5125]: Compiling Post-Auth-Type REJECT for attr Post-Auth-Type
Oct 28 16:36:20 kali freeradius[5125]: Compiling Post-Auth-Type Challenge for attr Post-Auth-Type
Oct 28 16:36:20 kali freeradius[5125]: Compiling Post-Auth-Type Client-Lost for attr Post-Auth-Type
Oct 28 16:36:20 kali freeradius[5125]: radiusd: ### Skipping IP addresses and Ports ###
Oct 28 16:36:20 kali freeradius[5125]: Configuration appears to be OK
Oct 28 16:36:20 kali systemd[1]: Started freeradius.service - FreeRADIUS multi-protocol policy server.
root@kali:/etc/freeradius/3.0#
```

Bien con esto terminamos la configuración del servidor de free radius.

Configuración en Ubuntu 18, usamos ssh para la conexión

```
WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED! @
#####
It is possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:WototoEWaffmJXnJVCjRrLtnrc9mLLDqt9TBmv91T84.
Please contact your system administrator.
Add correct host key in C:\Users\MSI CYBORG 14\.ssh\known_hosts to get rid of this message.
Offending ED25519 key in C:\Users\MSI CYBORG 14\.ssh\known_hosts:8
Host key for 192.168.1.7 has changed and you have requested strict checking.
Host key verification failed.
>cd C:\Users\MSI CYBORG 14\.ssh\known_hosts
Set-Location: A positional parameter cannot be found that accepts argument 'CYBORG'.
>cd "C:\Users\MSI CYBORG 14\.ssh\known_hosts"
Set-Location: Cannot find path 'C:\Users\MSI CYBORG 14\.ssh\known_hosts' because it does not exist.
>code "C:\Users\MSI CYBORG 14\.ssh\known_hosts"
>notepad "C:\Users\MSI CYBORG 14\.ssh\known_hosts"
>ssh k3vIn@192.168.1.7
The authenticity of host '192.168.1.7 (192.168.1.7)' can't be established.
ED25519 key fingerprint is SHA256:WototoEWaffmJXnJVCjRrLtnrc9mLLDqt9TBmv91T84.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.7' (ED25519) to the list of known hosts.
k3vIn@192.168.1.7's password:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-213-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue Oct 28 20:43:09 UTC 2025

System load:  0.0          Processes:    195
Usage of /:   30.2% of 18.5GB  Users logged in: 1
Memory usage: 6%           IP address for ens33: 192.168.1.7
Swap usage:   0%

41 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

New release '20.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Oct 28 20:39:32 2025
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

k3vIn@192.168.1.7:~$
```

Instalamos todo lo necesario


```
kali
ubuntu 18

Get:3 http://bo.archive.ubuntu.com/ubuntu bionic/main amd64 ssl-cert all 1.0.39 [17.0 kB]
Get:4 http://bo.archive.ubuntu.com/ubuntu bionic-updates/main amd64 freeradius-config amd64 3.0.16+dfsg-1ubuntu3.2 [150 kB]
Get:5 http://bo.archive.ubuntu.com/ubuntu bionic/main amd64 libtalloc2 amd64 2.1.10-2ubuntu1 [36.0 kB]
Get:6 http://bo.archive.ubuntu.com/ubuntu bionic-updates/main amd64 libfreeradius3 amd64 3.0.16+dfsg-1ubuntu3.2 [180 kB]
Get:7 http://bo.archive.ubuntu.com/ubuntu bionic-updates/main amd64 freeradius-utils amd64 3.0.16+dfsg-1ubuntu3.2 [88.5 kB]
Get:8 http://bo.archive.ubuntu.com/ubuntu bionic-updates/main amd64 libdbi-perl amd64 1.640-1ubuntu0.3 [724 kB]
Fetched 1,563 kB in 2s (946 kB/s)
Preconfiguring packages ...
Selecting previously unselected package freeradius-common.
(Reading database ... 67524 files and directories currently installed.)
Preparing to unpack .../0-freeradius-common_3.0.16+dfsg-1ubuntu3.2_all.deb ...
Unpacking freeradius-common (3.0.16+dfsg-1ubuntu3.2) ...
Selecting previously unselected package make.
Preparing to unpack .../1-make_4.1-9.1ubuntu1_amd64.deb ...
Unpacking make (4.1-9.1ubuntu1) ...
Selecting previously unselected package ssl-cert.
Preparing to unpack .../2-ssl-cert_1.0.39_all.deb ...
Unpacking ssl-cert (1.0.39) ...
Selecting previously unselected package freeradius-config.
Preparing to unpack .../3-freeradius-config_3.0.16+dfsg-1ubuntu3.2_amd64.deb ...
Unpacking freeradius-config (3.0.16+dfsg-1ubuntu3.2) ...
Selecting previously unselected package libtalloc2:amd64.
Preparing to unpack .../4-libtalloc2_2.1.10-2ubuntu1_amd64.deb ...
Unpacking libtalloc2:amd64 (2.1.10-2ubuntu1) ...
Selecting previously unselected package libfreeradius3.
Preparing to unpack .../5-libfreeradius3_3.0.16+dfsg-1ubuntu3.2_amd64.deb ...
Unpacking libfreeradius3 (3.0.16+dfsg-1ubuntu3.2) ...
Selecting previously unselected package freeradius-utils.
Preparing to unpack .../6-freeradius-utils_3.0.16+dfsg-1ubuntu3.2_amd64.deb ...
Unpacking freeradius-utils (3.0.16+dfsg-1ubuntu3.2) ...
Selecting previously unselected package libdbi-perl.
Preparing to unpack .../7-libdbi-perl_1.640-1ubuntu0.3_amd64.deb ...
Unpacking libdbi-perl (1.640-1ubuntu0.3) ...
Setting up make (4.1-9.1ubuntu1) ...
Setting up freeradius-common (3.0.16+dfsg-1ubuntu3.2) ...
Adding user freerad to group shadow
Setting up ssl-cert (1.0.39) ...
Setting up freeradius-config (3.0.16+dfsg-1ubuntu3.2) ...
Adding user freerad to group ssl-cert
Generating DH parameters, 1024 bit long safe prime, generator 2
This is going to take a long time
.....+-----+
Setting up libtalloc2:amd64 (2.1.10-2ubuntu1) ...
Setting up libdbi-perl (1.640-1ubuntu0.3) ...
Setting up libfreeradius3 (3.0.16+dfsg-1ubuntu3.2) ...
Setting up freeradius-utils (3.0.16+dfsg-1ubuntu3.2) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
Processing triggers for libc-bin (2.27-3ubuntu1.5) ...
k3v1n@h4ck:~$
```

Probamos la conexión con: radtest "k3v1n h4ck" 12345 192.168.1.10 1812 sistemas

La petición es rechazada, probamos que esta fallando

Pienso que falla aquí: en clients.conf

```
GNU nano 8.6
clients.conf *

# You can have as many per-socket client lists as you have "listen"
# sections, or you can re-use a list among multiple "listen" sections.
#
# Un-comment this section, and edit a "listen" section to add:
# "clients = per_socket_clients". That IP address/port combination
# will then accept ONLY the clients listed in this section.
#
# There are additional considerations when using clients from SQL.
#
# A client can be link to a virtual server via modules such as SQL.
# This link is done via the following process:
#
# If there is no listener in a virtual server, SQL clients are added
# to the global list for that virtual server.
#
# If there is a listener, and the first listener does not have a
# "clients=..." configuration item, SQL clients are added to the
# global list.
#
# If there is a listener, and the first one does have a "clients=..."
# configuration item, SQL clients are added to that list. The client
# { ... } configured in that list are also added for that listener.
#
# The only issue is if you have multiple listeners in a virtual
# server, each with a different client list, then the SQL clients are
# added only to the first listener.
#
#clients_per_socket_clients {
#   client socket_client {
#     ipaddr = 192.0.2.4
#     secret = testing123
#   }
#}

client 192.168.1.7 {
    secret = sistemas
    shortname = router1
    nasstype = cisco
}

client 0.0.0.0/0 {
    ipaddr = 0.0.0.0/0
    secret = sistemas
    shortname = windows-client
}

# Help
# Exit
# Write Out
# Read File
# Where Is
# Replace
# Cut
# Paste
# Execute
# Justify
# Location
# Go To Line
# M-U
# Undo
# M-E
# Redo
# M-A
# Set Mark
# M-6
# Copy
# M-B
# To Bracket
# M-7
# Where Was
# M-F
# Previous
# M-N
# Next
# Back
# Forward
```

Aclaracion: Antes la ip es 192.168.1.10, pero la documentación dice que esta ip tiene que ser la ip del host que tratara de autenticarse con freeradius. Pero esta ip es del servidor de freeradius.

Probamos nuevamente

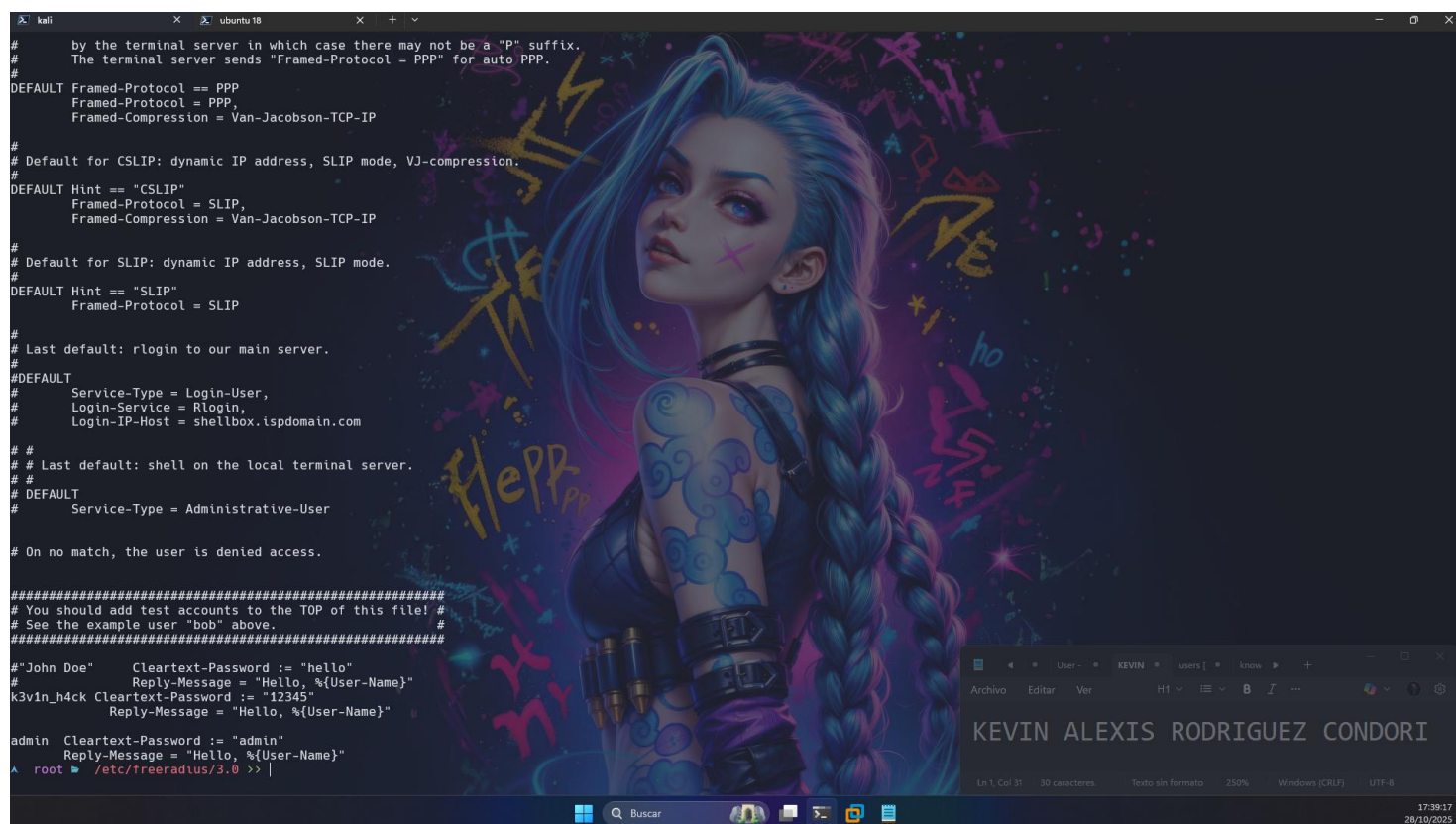
Sigue sin funcionar, entonces al hacer muchas pruebas descubrí que con “radtest admin admin 192.168.1.10 18120 sistemas”, funciona normal pero con radtest "k3v1n h4ck" 12345 192.168.1.10 18120 sistemas, no funciona.

Configuramos nuevamente a
k3v1n_h4ck Cleartext-Password := "12345"

Reply-Message = "Hello, %{User-Name}"

admin Cleartext-Password := "admin"

Reply-Message = "Hello, %{User-Name}"



```
# by the terminal server in which case there may not be a "P" suffix.
# The terminal server sends "Framed-Protocol = PPP" for auto PPP.
#
DEFAULT Framed-Protocol == PPP
Framed-Protocol = PPP,
Framed-Compression = Van-Jacobson-TCP-IP

#
# Default for CSLIP: dynamic IP address, SLIP mode, VJ-compression.
#
DEFAULT Hint == "CSLIP"
Framed-Protocol = SLIP,
Framed-Compression = Van-Jacobson-TCP-IP

#
# Default for SLIP: dynamic IP address, SLIP mode.
#
DEFAULT Hint == "SLIP"
Framed-Protocol = SLIP

#
# Last default: rlogin to our main server.
#
#DEFAULT
#
#   Service-Type = Login-User,
#   Login-Service = Rlogin,
#   Login-IP-Host = shellbox.ispdomain.com
#
#
# Last default: shell on the local terminal server.
#
#DEFAULT
#
#   Service-Type = Administrative-User

#
# On no match, the user is denied access.
#
#####
# You should add test accounts to the TOP of this file! #
# See the example user "bob" above.
#####

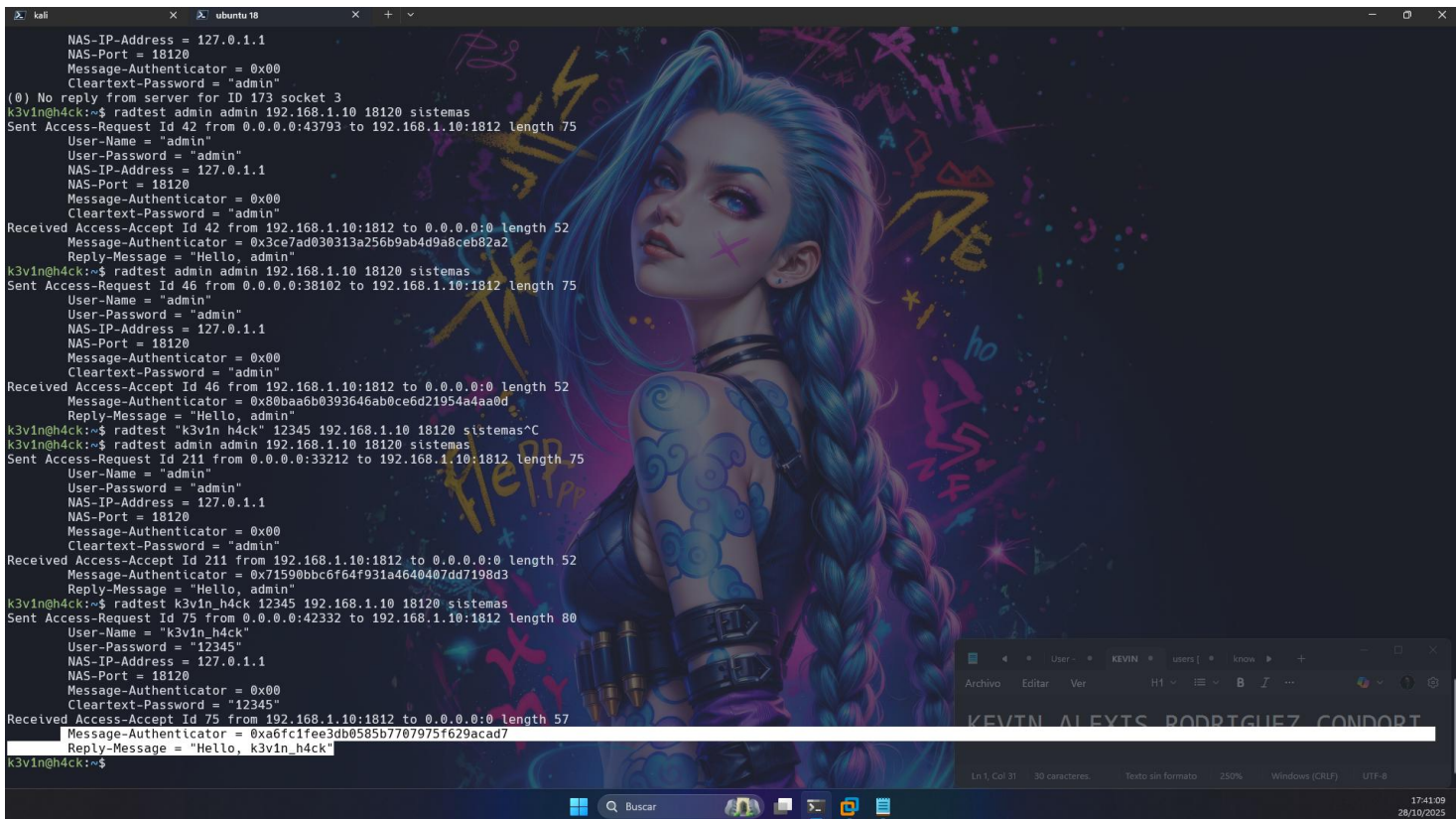
# "John Doe"      Cleartext-Password := "hello"
#                  Reply-Message = "Hello, %{User-Name}"
k3v1n_h4ck Cleartext-Password := "12345"
#                  Reply-Message = "Hello, %{User-Name}"

admin Cleartext-Password := "admin"
#                  Reply-Message = "Hello, %{User-Name}"

root # /etc/freeradius/3.0 >> |
```

Entonces probamos nuevamente con esto:

radtest k3v1n_h4ck 12345 192.168.1.10 18120 sistemas



Entonces todo va bien.

Entonces la observación anterior no es como tal un error, pero se considera una mala práctica, esto debido a que en la línea de configuración:

```
client 0.0.0.0/0 {  
    ipaddr      = 0.0.0.0/0  
    secret      = sistemas  
    shortname   = windows-client  
}
```

Esto acepta cualquier origen, útil para pruebas, pero **no recomendado en producción**.

La configuración correcta en el archivo de configuración debería de ser:

```
client <ip> {  
    secret      = sistemas  
    shortname   = router1  
    nastype     = cisco  
}
```

Donde la ip debería de ser la del host de origen, porque la forma en como funciona el servicio es:

- ❖ En clients.conf, FreeRADIUS valida la **IP de origen** de la petición.
- ❖ Si no encuentra un bloque client que coincida con la IP de origen (192.168.1.7), rechaza la solicitud con *Access-Reject*.

Pero en este caso la ip del cliente configurado es la ip del servidor y gracias al aliente wildard funciona.

Pruebas:

Comentamos la línea del cliente wilcard:

```
#####
#
# Per-socket client lists. The configuration entries are exactly
# the same as above, but they are nested inside of a section.
#
# You can have as many per-socket client lists as you have "listen"
# sections, or you can re-use a list among multiple "listen" sections.
#
# Un-comment this section, and edit a "listen" section to add:
# "clients = per_socket_clients". That IP address/port combination
# will then accept ONLY the clients listed in this section.
#
# There are additional considerations when using clients from SQL.
#
# A client can be link to a virtual server via modules such as SQL.
# This link is done via the following process:
#
# If there is no listener in a virtual server, SQL clients are added
# to the global list for that virtual server.
#
# If there is a listener, and the first listener does not have a
# "clients=..." configuration item, SQL clients are added to the
# global list.
#
# If there is a listener, and the first one does have a "clients=..."
# configuration item, SQL clients are added to that list. The client
# { ... } configured in that list are also added for that listener.
#
# The only issue is if you have multiple listeners in a virtual
# server, each with a different client list, then the SQL clients are
# added only to the first listener.
#
#clients per_socket_clients {
#    client socket_client {
#        ipaddr = 192.0.2.4
#        secret = testing123
#    }
#}

client 192.168.1.10 {
    secret = sistemas
    shortname = router1
    nastype = cisco
}

#client 0.0.0.0/0 {
#    ipaddr = 0.0.0.0/0
#    secret = sistemas
#    shortname = windows-client
#}

root@kali: /etc/freeradius/3.0 >> |
```

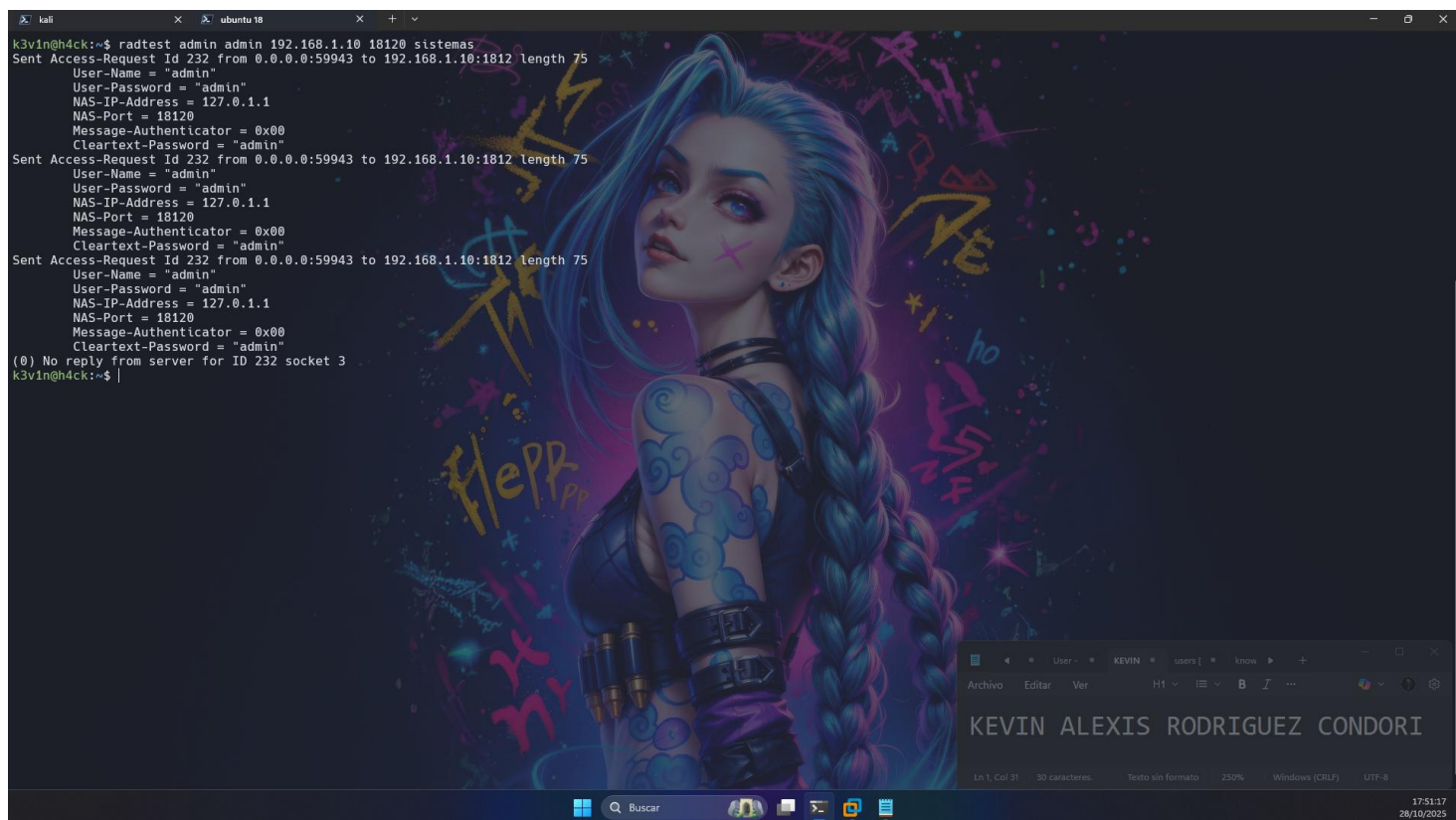
Intentamos hacer la petición de autenticación:

Observaremos que intenta una y otra vez pero al final no lo logra

```
k3vin@h4ck:~$ radtest k3vin_h4ck 12345 192.168.1.10 18120 sistemas
Sent Access-Request Id 146 from 0.0.0.0:43012 to 192.168.1.10:1812 length 80
  User-Name = "k3vin_h4ck"
  User-Password = "12345"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "12345"
Sent Access-Request Id 146 from 0.0.0.0:43012 to 192.168.1.10:1812 length 80
  User-Name = "k3vin_h4ck"
  User-Password = "12345"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "12345"
Sent Access-Request Id 146 from 0.0.0.0:43012 to 192.168.1.10:1812 length 80
  User-Name = "k3vin_h4ck"
  User-Password = "12345"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "12345"
(0) No reply from server for ID 146 socket 3
k3vin@h4ck:~$ |
```


Lo mismo pasa con admin

```
kali k3vin@h4ck:~$ radtest admin admin 192.168.1.10 18120 sistemas
Sent Access-Request Id 232 from 0.0.0.0:59943 to 192.168.1.10:1812 length 75
  User-Name = "admin"
  User-Password = "admin"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "admin"
Sent Access-Request Id 232 from 0.0.0.0:59943 to 192.168.1.10:1812 length 75
  User-Name = "admin"
  User-Password = "admin"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "admin"
Sent Access-Request Id 232 from 0.0.0.0:59943 to 192.168.1.10:1812 length 75
  User-Name = "admin"
  User-Password = "admin"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "admin"
(0) No reply from server for ID 232 socket 3
k3vin@h4ck:~$
```



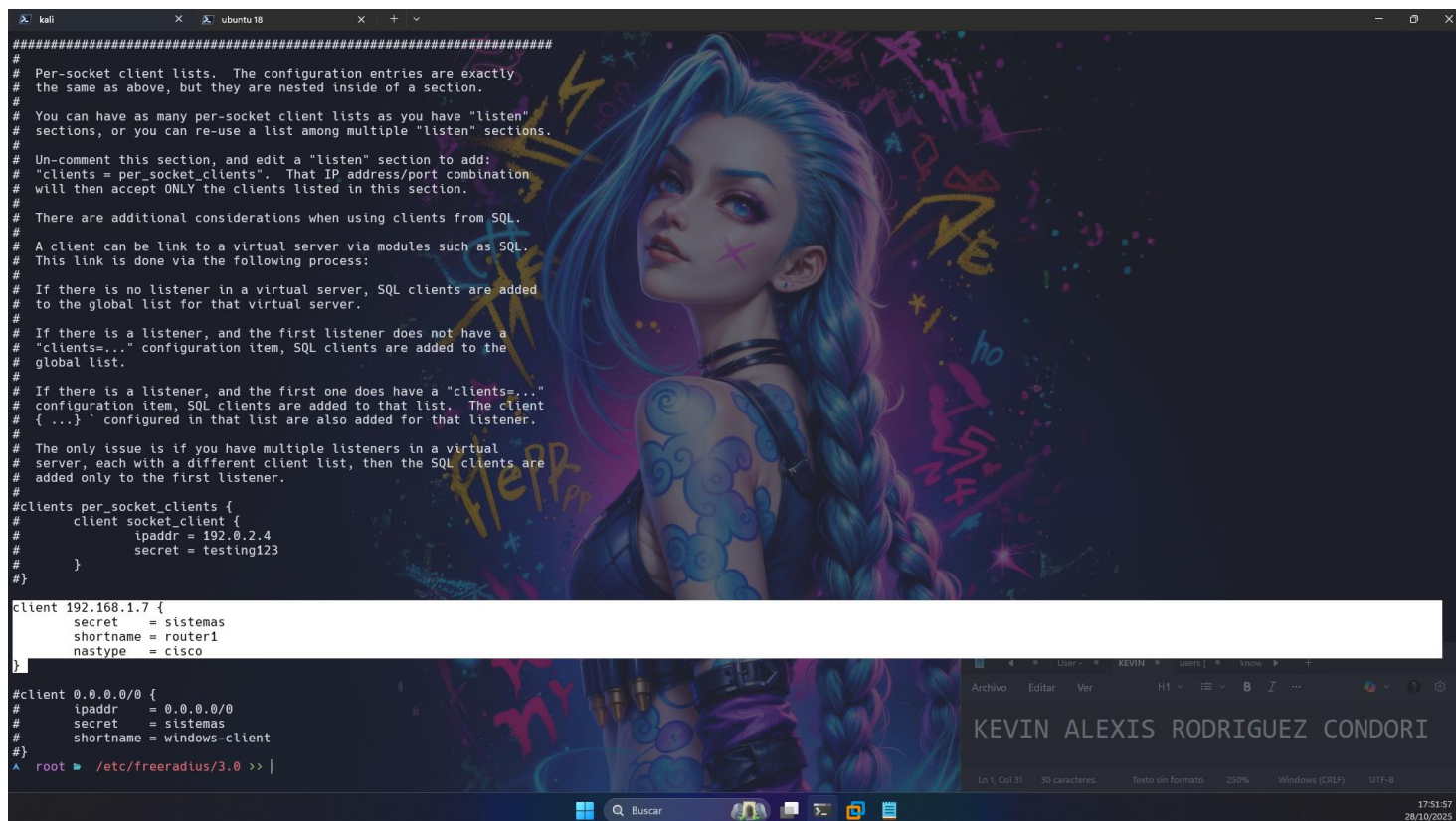
Pero si cambiamos la ip de origen en clients.conf, esto funciona

```
#####
#
# Per-socket client lists. The configuration entries are exactly
# the same as above, but they are nested inside of a section.
#
# You can have as many per-socket client lists as you have "listen"
# sections, or you can re-use a list among multiple "listen" sections.
#
# Un-comment this section, and edit a "listen" section to add:
# "clients = per_socket_clients". That IP address/port combination
# will then accept ONLY the clients listed in this section.
#
# There are additional considerations when using clients from SQL.
#
# A client can be link to a virtual server via modules such as SQL.
# This link is done via the following process:
#
# If there is no listener in a virtual server, SQL clients are added
# to the global list for that virtual server.
#
# If there is a listener, and the first listener does not have a
# "clients=..." configuration item, SQL clients are added to the
# global list.
#
# If there is a listener, and the first one does have a "clients=..."
# configuration item, SQL clients are added to that list. The client
# { ... } configured in that list are also added for that listener.
#
# The only issue is if you have multiple listeners in a virtual
# server, each with a different client list, then the SQL clients are
# added only to the first listener.
#
#clients per_socket_clients {
#
#   client socket_client {
#     ipaddr = 192.0.2.4
#     secret = testing123
#   }
#}

client 192.168.1.7 {
  secret = sistemas
  shortname = router1
  nasstype = cisco
}

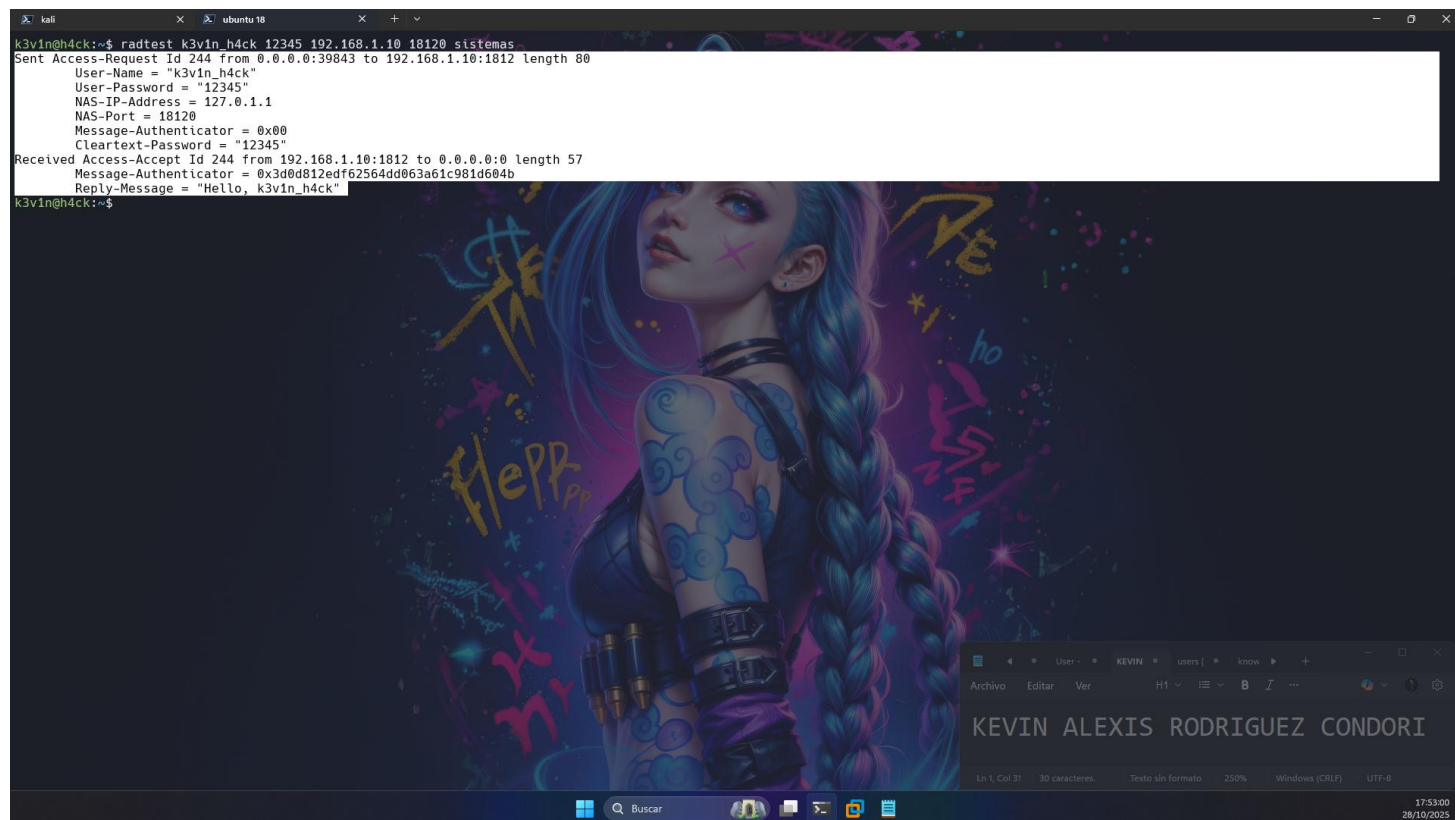
#client 0.0.0.0/0 {
#  ipaddr = 0.0.0.0/0
#  secret = sistemas
#  shortname = windows-client
#}

^ root@ /etc/freeradius/3.0 >> |
```



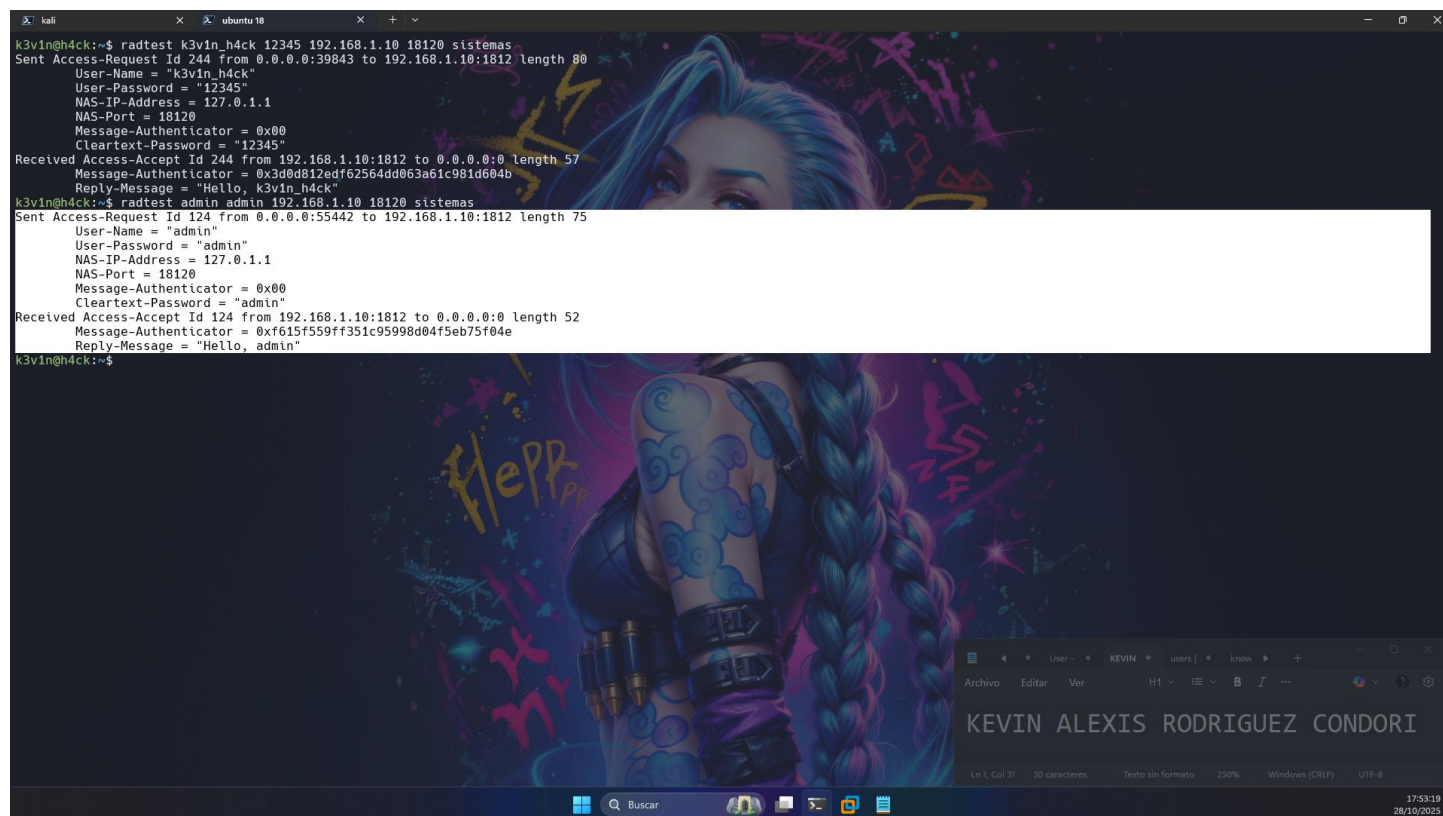
Ahora probamos nuevamente:

```
kali k3vin@h4ck:~$ radtest k3vin_h4ck 12345 192.168.1.10 18120 sistemas
Sent Access-Request Id 244 from 0.0.0.0:39843 to 192.168.1.10:1812 length 80
  User-Name = "k3vin_h4ck"
  User-Password = "12345"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "12345"
Received Access-Accept Id 244 from 192.168.1.10:1812 to 0.0.0.0:0 length 57
  Message-Authenticator = 0x3d0d812edf62564dd063a61c981d604b
  Reply-Message = "Hello, k3vin_h4ck"
k3vin@h4ck:~$
```



Admin

```
kali k3vin@h4ck:~$ radtest k3vin_h4ck 12345 192.168.1.10 18120 sistemas
Sent Access-Request Id 244 from 0.0.0.0:39843 to 192.168.1.10:1812 length 80
  User-Name = "k3vin_h4ck"
  User-Password = "12345"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "12345"
Received Access-Accept Id 244 from 192.168.1.10:1812 to 0.0.0.0:0 length 57
  Message-Authenticator = 0x3d0d812edf62564dd063a61c981d604b
  Reply-Message = "Hello, k3vin_h4ck"
k3vin@h4ck:~$ radtest admin admin 192.168.1.10 18120 sistemas
Sent Access-Request Id 124 from 0.0.0.0:55442 to 192.168.1.10:1812 length 75
  User-Name = "admin"
  User-Password = "admin"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "admin"
Received Access-Accept Id 124 from 192.168.1.10:1812 to 0.0.0.0:0 length 52
  Message-Authenticator = 0xf615f559ff351c95998d04f5eb75f04e
  Reply-Message = "Hello, admin"
k3vin@h4ck:~$
```

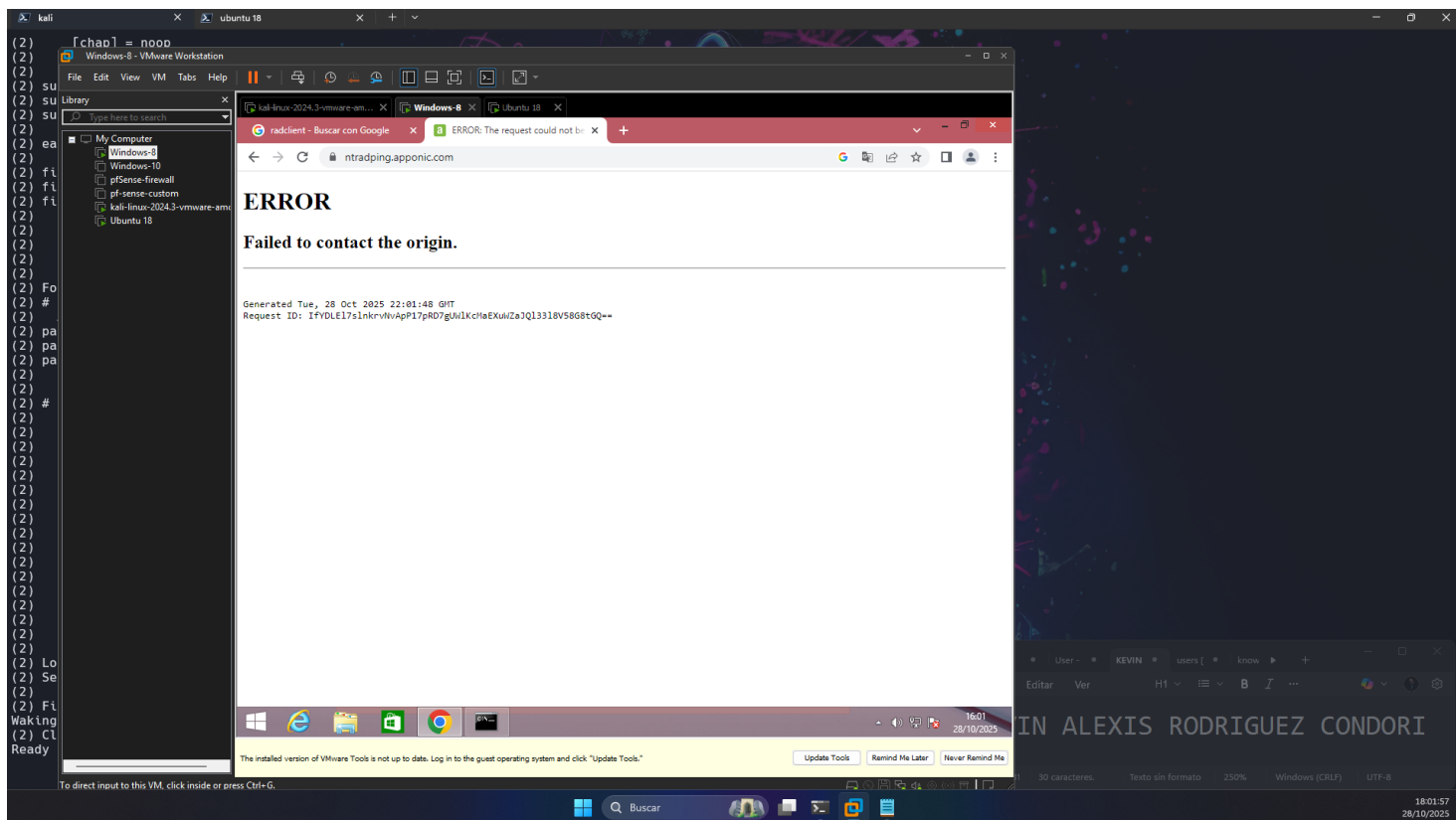


Entonces esta configuración limita a los host de origen, igual se le puede especificar rangos de IPs.

Ahora hagamos las pruebas en Windows 8

No se tiene instalado “NTRadping”

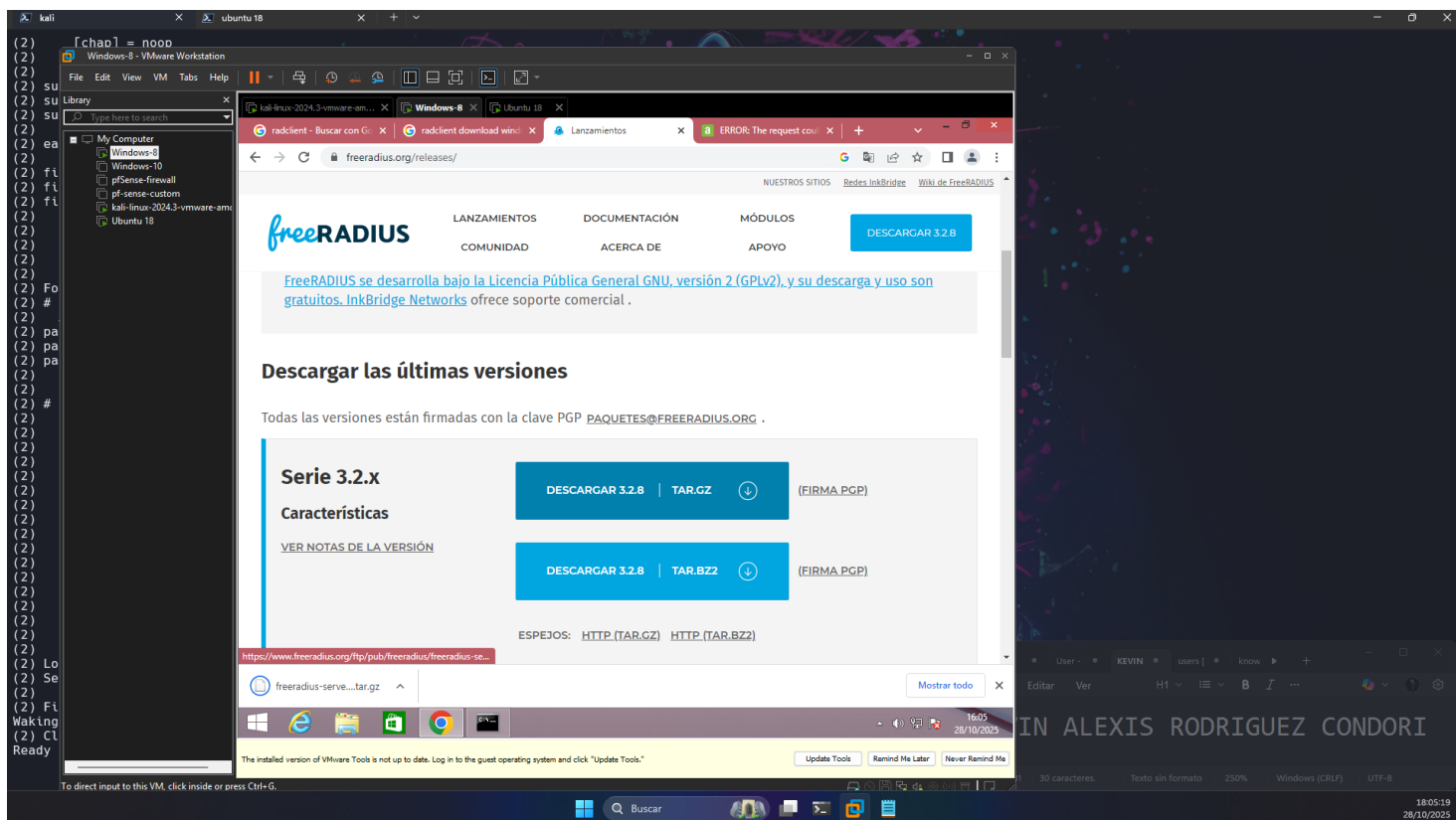
Instalación:



No se puede descargar

Buscamos alternativas: radclient

<https://www.freeradius.org/releases/>

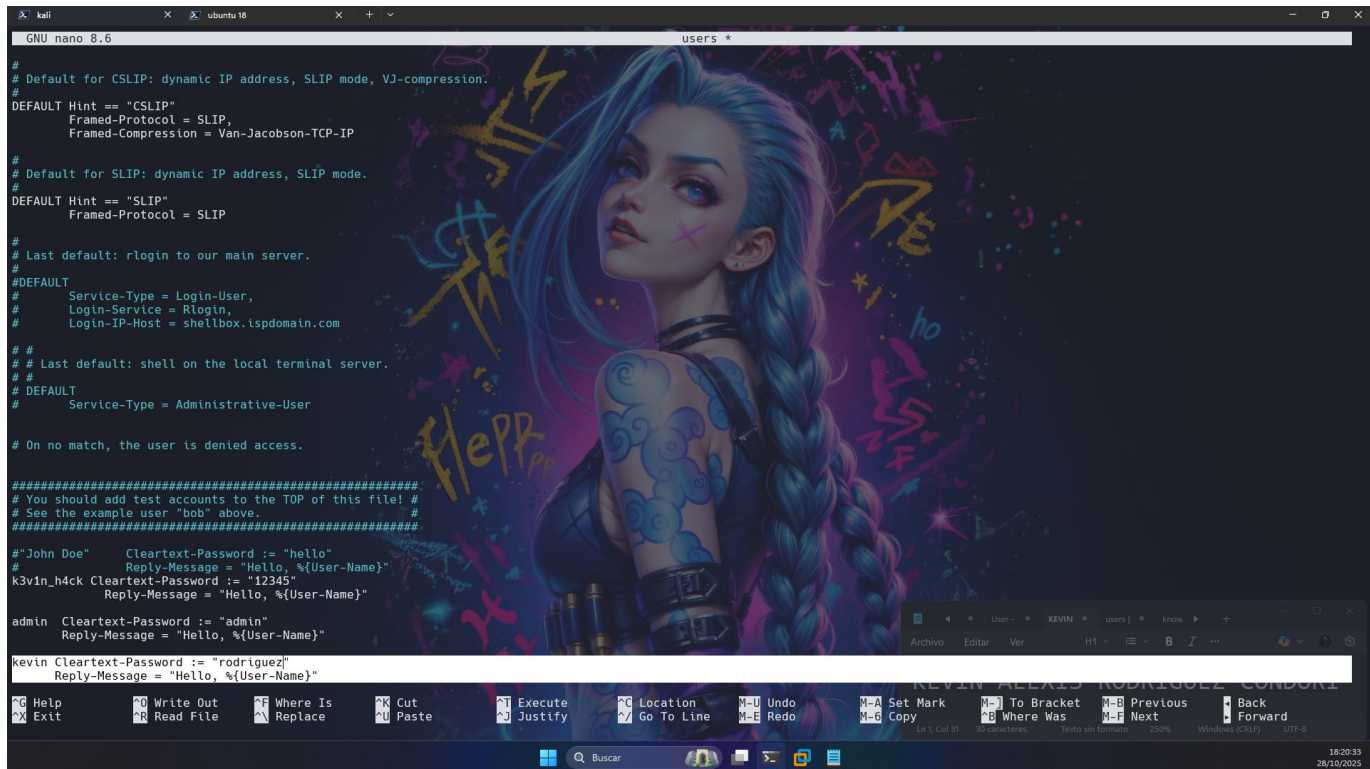


No se logra la instalación de ninguno de los clientes

Evaluación

1. Con que comando se puede ver los logs en tiempo real en el servidor radius.
freeradius -X o sudo tail -f /var/log/freeradius/radius.log
2. Crear un nuevo usuario e logearse tanto desde Linux como desde Windows e indique que datos puede observar en estos logs. Al crear el usuario ponga su nombre como usuario y su apellido como contraseña.

Creación del nuevo usuario



```
GNU nano 8.6 users *
# Default for CSLIP: dynamic IP address, SLIP mode, VJ-compression.
#
DEFAULT Hint == "CSLIP"
Framed-Protocol = SLIP,
Framed-Compression = Van-Jacobson-TCP-IP

# Default for SLIP: dynamic IP address, SLIP mode.
#
DEFAULT Hint == "SLIP"
Framed-Protocol = SLIP

#
# Last default: rlogin to our main server.
#
#DEFAULT
#
#   Service-Type = Login-User,
#   Login-Service = Rlogin,
#   Login-IP-Host = shellbox.ispdomain.com

#
# Last default: shell on the local terminal server.
#
#DEFAULT
#
#   Service-Type = Administrative-User

# On no match, the user is denied access.

#####
# You should add test accounts to the TOP of this file! #
# See the example user "bob" above. #
#####

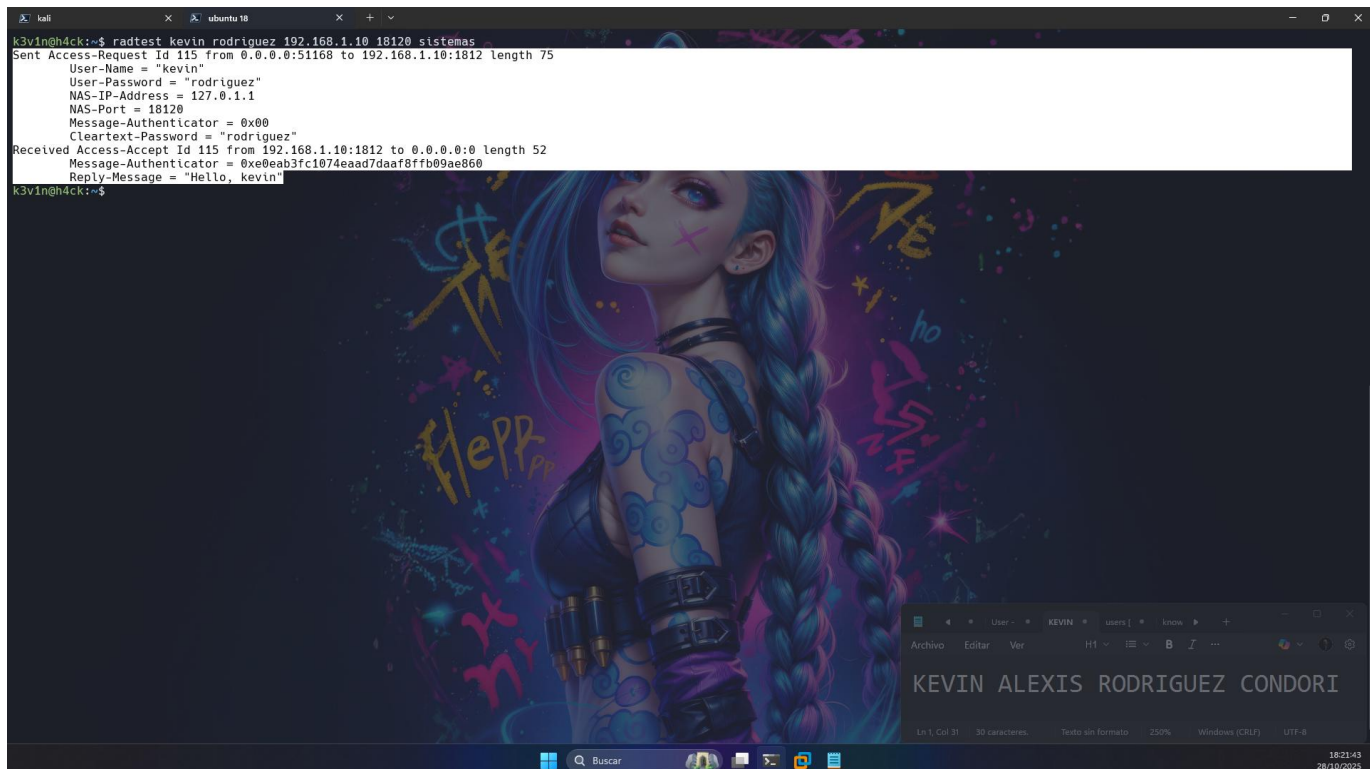
#"John Doe"   Cleartext-Password := "hello"
#             Reply-Message = "Hello, %{User-Name}"
k3v1n_h4ck Cleartext-Password := "12345"
#             Reply-Message = "Hello, %{User-Name}"

admin Cleartext-Password := "admin"
#             Reply-Message = "Hello, %{User-Name}"

kevin Cleartext-Password := "rodriguez"
#             Reply-Message = "Hello, %{User-Name}"

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward
```

Probamos la conexión de Linux



```
k3v1n@h4ck:~$ radtest kevin rodriguez 192.168.1.10 18120 sistemas
Sent Access-Request Id 115 from 0.0.0.0:51168 to 192.168.1.10:1812 Length 75
  User-Name = "kevin"
  User-Password = "rodriguez"
  NAS-IP-Address = 127.0.1.1
  NAS-Port = 18120
  Message-Authenticator = 0x00
  Cleartext-Password = "rodriguez"
Received Access-Accept Id 115 from 192.168.1.10:1812 to 0.0.0.0:0 Length 52
  Message-Authenticator = 0xe0eab3fc1074ead7daaf8ffb09ae060
  Reply-Message = "Hello, kevin"
k3v1n@h4ck:~$
```

Lo que se puede observar en los logs es lo siguiente

